

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0282

Garnet Mesa 138 kV
October 01, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0282. The Interconnection Customer proposes a 76.2 MW solar photovoltaic facility interconnecting at the Garnet Mesa 138 kV substation in the Pennsylvania area. The request was received on November 03, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 206 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,687,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of ALLEGHENY ENERGY UNITS 3 4 & 5 / SOUTHEAST VICKSBURG 500.0 kV, the HOUSEVILLE / FRALEY 138.0 kV Line loads to 190.3% of its 179 MVA rating. Project DFAX: 9.7%.

For the loss of SOUTHEAST VICKSBURG / ALLEGHENY ENERGY UNITS 3 4 & 5 500.0 kV, the FRALEY SUBSTATION / YUCCA 138.0 kV loads to 184.4% of its 180 MVA rating. Project DFAX: 9.7%.

The HOUSEVILLE / VA HOSPITAL 138.0 kV is overloaded to 181.2% of its 190 MVA emergency rating for the loss of the ALLEGHENY ENERGY UNITS 3 4 & 5 / SOUTHEAST VICKSBURG 500.0 kV. The project contributes 9.7% DFAX (7.4 MW) to this overload.

The SUGAR HILL / VA HOSPITAL 138.0 kV Line is overloaded to 171.5% of its 204 MVA emergency rating for the loss of the ALLEGHENY ENERGY UNITS 3 4 & 5 / SOUTHEAST VICKSBURG 500.0 kV L. The project contributes 9.7% DFAX (7.4 MW) to this overload.

SUGAR HILL / ALLEGHENY ENERGY UNITS 3 4 & 5 345.0 kV: 102.0% loading (1,163 MVA) following the outage of the HILLTOP / PINEY CRK PROJECT 500.0 kV; DFAX = 18.1%.

For the loss of ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV/1 kV TX, the BURWELL 500-345 kV XFMR loads to 101.0% of its 692 MVA rating. Project DFAX: 6.7%.

The MARICAO 6301 / STA J 138.0 kV Line is overloaded to 100.3% of its 138 MVA emergency rating for the loss of the REGAL / YUCCA 138.0 kV Line. The project contributes 6.1% DFAX (4.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FRALEY / HOUSEVILLE 138.0 kV	Rebuild line with higher-capacity conductor	\$4,207,000	100% (\$4,207,000)
FRALEY SUBSTATION / YUCCA 138.0 kV	Rebuild line with higher-capacity conductor	\$1,931,000	100% (\$1,931,000)
HOUSEVILLE / VA HOSPITAL 138.0 kV	Rebuild line with higher-capacity conductor	\$2,606,000	100% (\$2,606,000)
SUGAR HILL / VA HOSPITAL 138.0 kV	Rebuild line with higher-capacity conductor	\$6,648,000	100% (\$6,648,000)
SUGAR HILL / ALLEGHENY ENERGY UNITS 3 4 & 5 345.0 kV	Reconductor line	\$7,930,000	3% (\$258,000)
BURWELL 500-345 kV Transformer	Replace transformer with higher-rated unit	\$31,541,000	2% (\$518,000)
MARICAO 6301 / STATION J 138.0 kV line	Reconductor line	\$519,000	100% (\$519,000)
	Total allocated to project		\$16,687,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.